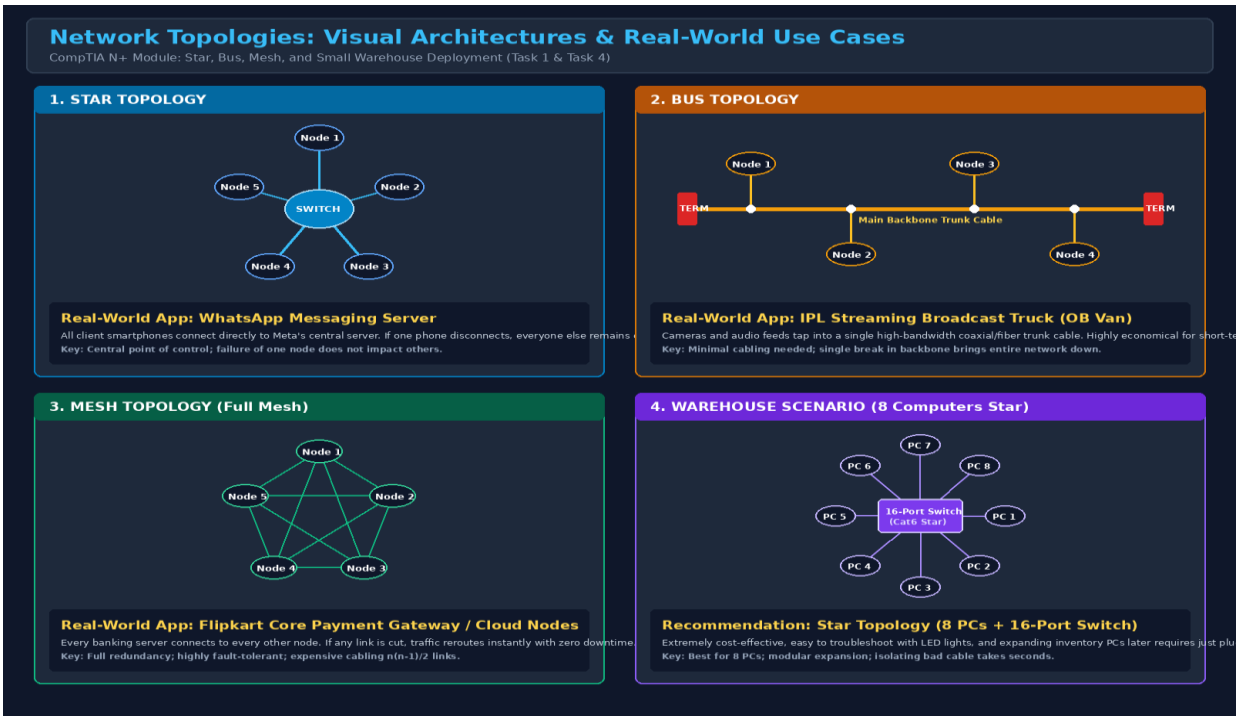
Network Topologies

1. Network Topology Diagrams & Real-World Services

An annotated diagram illustrating Star, Bus, Mesh, and the Warehouse Star network is provided below:



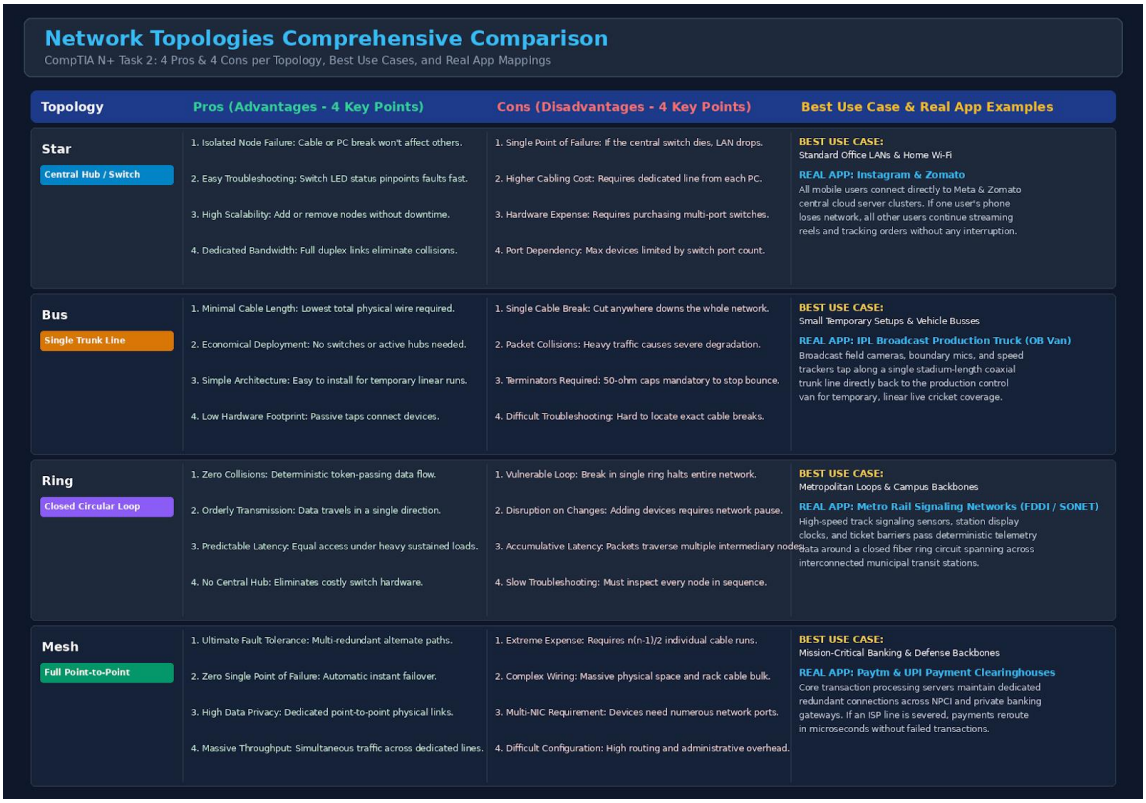
Real-World App & Service Mapping:

- **Star Topology — WhatsApp Messaging Architecture:**
    - *Why:* All smartphones communicate through Meta's central messaging servers acting as the central hub. If one user's phone goes offline or loses signal, the rest of the messaging network operates without disruption.
- **Bus Topology — IPL Live Streaming Broadcast Truck (Outside Broadcasting Van):**
    - *Why:* Field cameras, scoreboards, and commentary audio feeds along the stadium boundary tap into a single long coaxial or fiber trunk cable running back to the production truck, minimizing cable clutter for temporary live sports coverage.
- **Mesh Topology — Flipkart Core Payment Gateway / Banking Infrastructure:**

- *Why:* Critical transaction and database servers are interconnected with multiple redundant routes. If one network link or server fails during a flash sale, traffic reroutes in microseconds to prevent transaction timeouts.

2. Comprehensive Comparison Table: Star, Bus, Ring, and Mesh

Topology Pros (Advantages) Cons (Disadvantages) Best Use Case App / Service Example

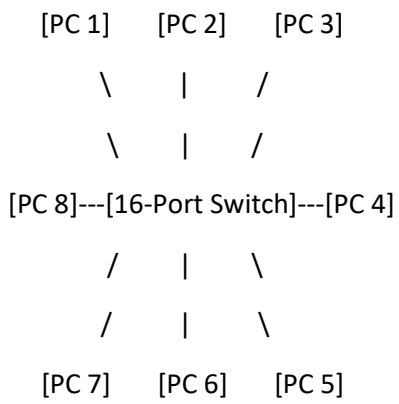


3. Multiplayer Mobile Game Server Topology (PUBG / Free Fire)

- **Recommended Architecture: Hybrid Star-Mesh Topology** *(Clients connect via Star to Regional Edge Servers; Edge Servers interconnect in a Mesh back to the Central Game Engine)*.
- **Justification (Speed, Reliability, & Cost):**
    - **Speed (Latency/Ping):** A Star topology between players and regional edge servers minimizes packet hops, keeping UDP game telemetry latency under 20–30 ms20\text{–}30\text{ ms}20–30 ms for real-time hit registration.
    - **Reliability:** The core database, matchmaking, and player inventory servers are connected in a fault-tolerant **Mesh topology**, ensuring that if an AWS/Google Cloud link fails, matchmaking and anti-cheat validations continue across redundant routes.
    - **Cost:** Pure mesh to every mobile player is technically impossible; this hybrid design concentrates expensive redundant mesh connections strictly in the cloud data centers while leveraging low-cost, scalable star connections for millions of mobile devices.

4. Flipkart-Style Warehouse Scenario (8 Computers Deployment)

- **Recommended Topology: Star Topology** (8 Desktop Workstations connected to a central **16-Port Gigabit Switch** using Cat6 UTP cables).



- **Reasoning:**

1. **Cost-Effective & Simple:** Standard Cat6 patch cables and an unmanaged 16-port Gigabit switch keep installation well within a small warehouse budget.
2. **Ease of Troubleshooting:** Each computer has an isolated, dedicated cable run. If a forklift severs the cable to PC 3, only that inventory station drops offline while the other 7 order-processing PCs continue operating uninterrupted; the blinking port LED on the switch immediately identifies which line is down.
3. **Future Expansion:** Using a 16-port switch leaves 8 spare Gigabit ports available to connect barcode label printers, handheld Wi-Fi access points, or additional packing desks without re-architecting the network.